

Cyber and Information Security

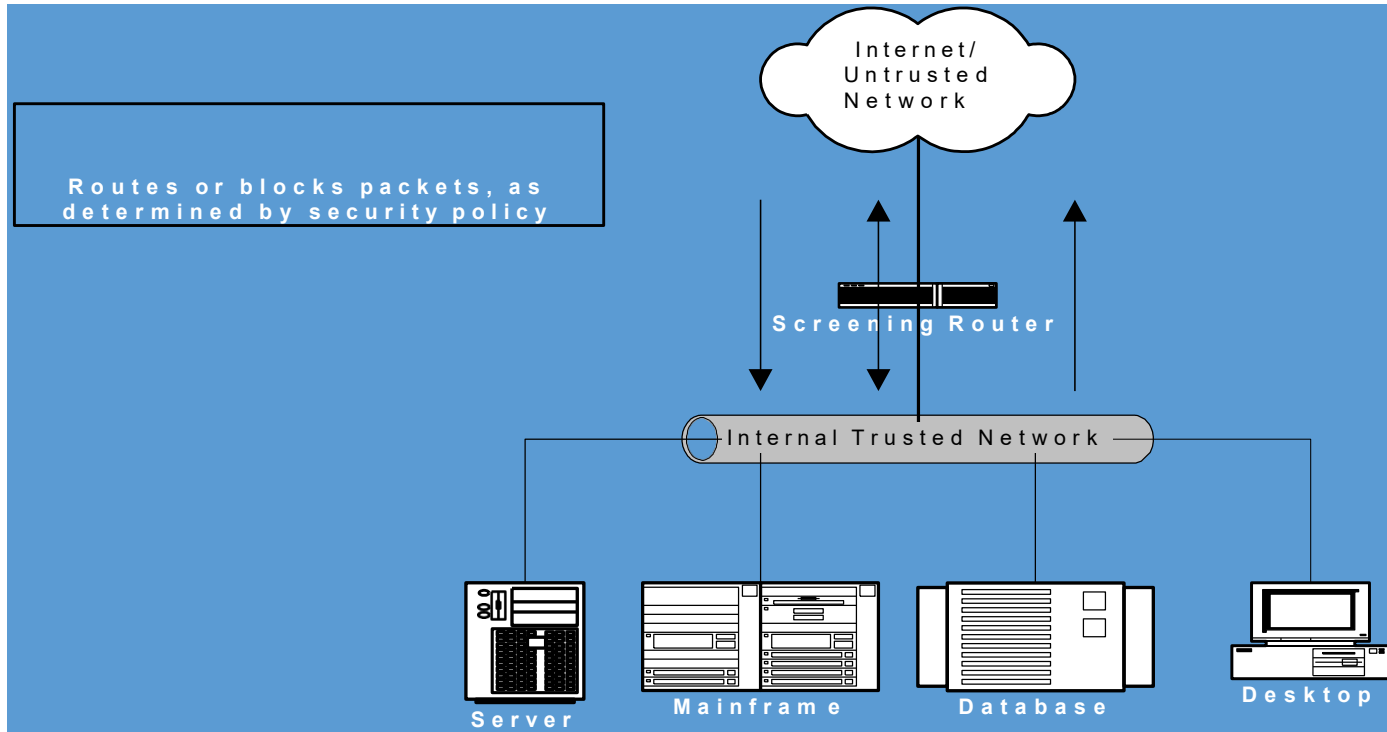
Lecture 7.



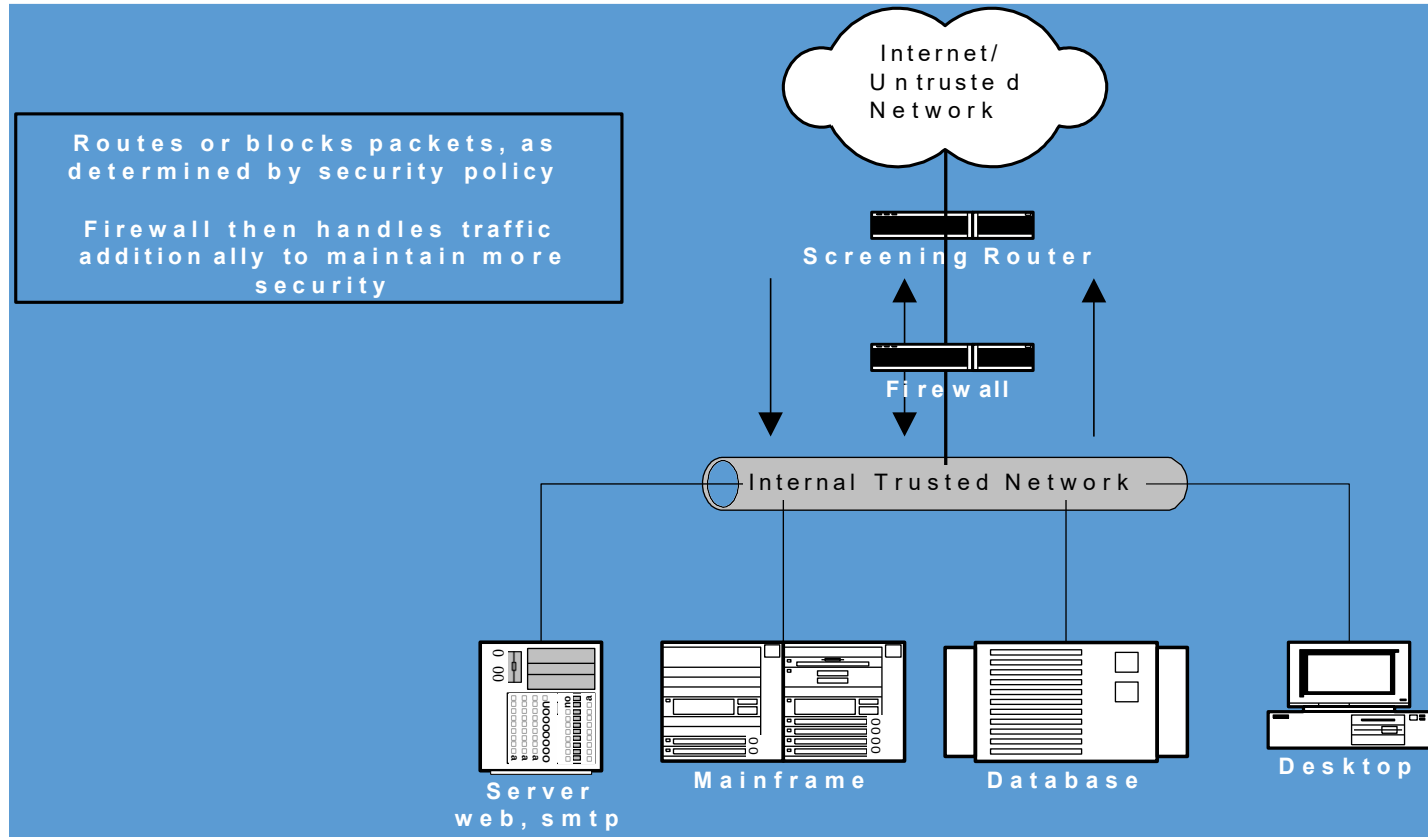
NETWORK FIREWALL ARCHITECTURES

- Screening Router
- Simple Firewall
- Multi-Legged firewall

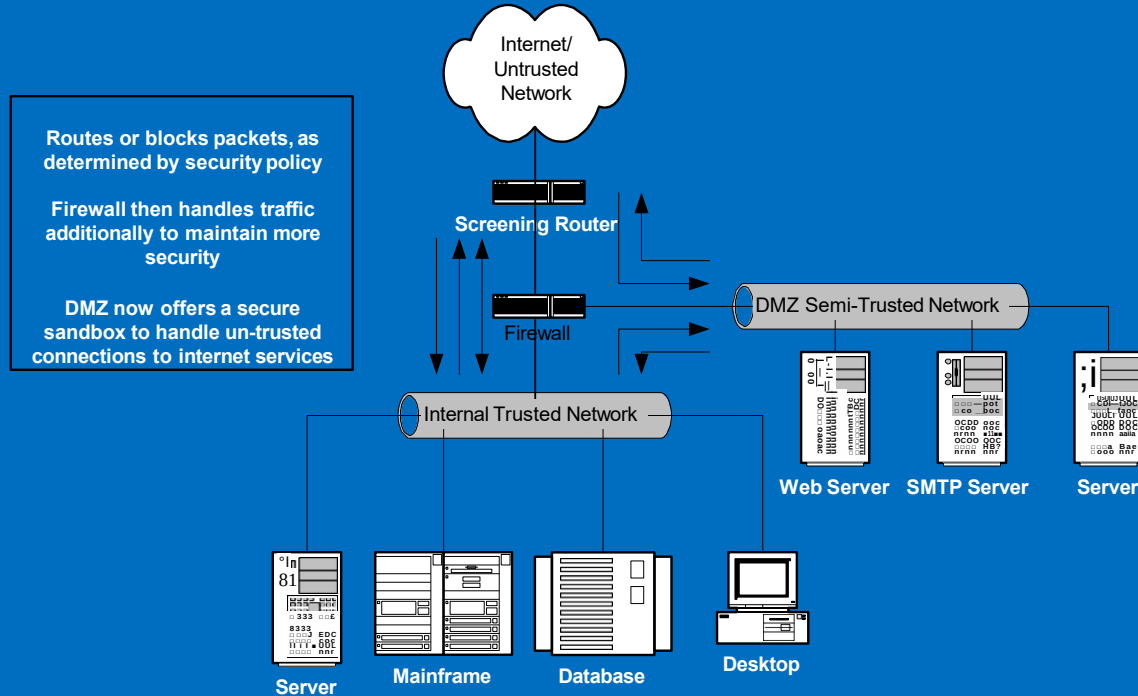
SCREENING ROUTER



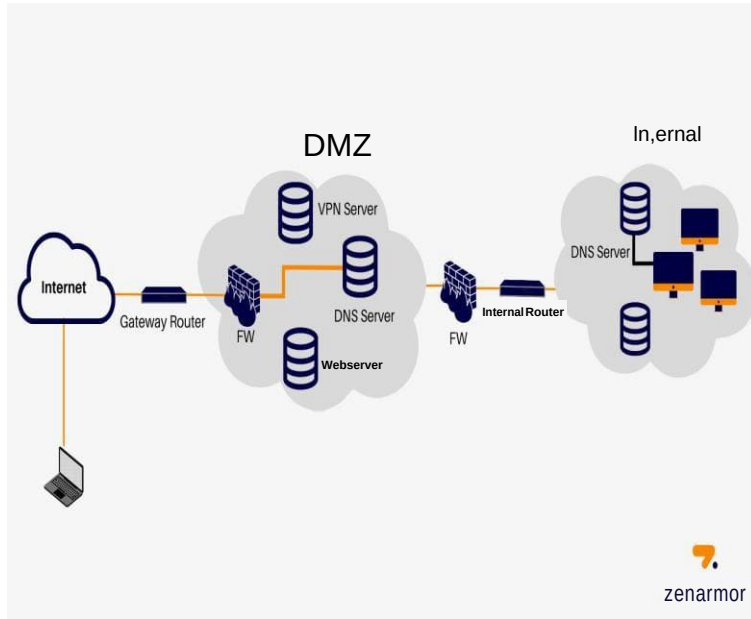
SIMPLE FIREWALL



MULTI-LEGGED FIREWALL



DMZ SANDBOX



- The DMZ is like a separate sandbox for external facing machines prevent access from your private network.

FIREWALL CHARACTERISTICS

Four general techniques:

1. Service control

- Determines the types of Internet services that can be accessed, inbound or outbound

2. Direction control

- Determines the direction in which particular service requests are allowed to flow

3. User control

- Controls access to a service according to which user is attempting to access it

4. Behavior control

- Controls how particular services are used (e.g. filter e-mail)

Firewall Limitations

- cannot protect from attacks bypassing it
- cannot protect against internal threats
- cannot protect against access via WLAN

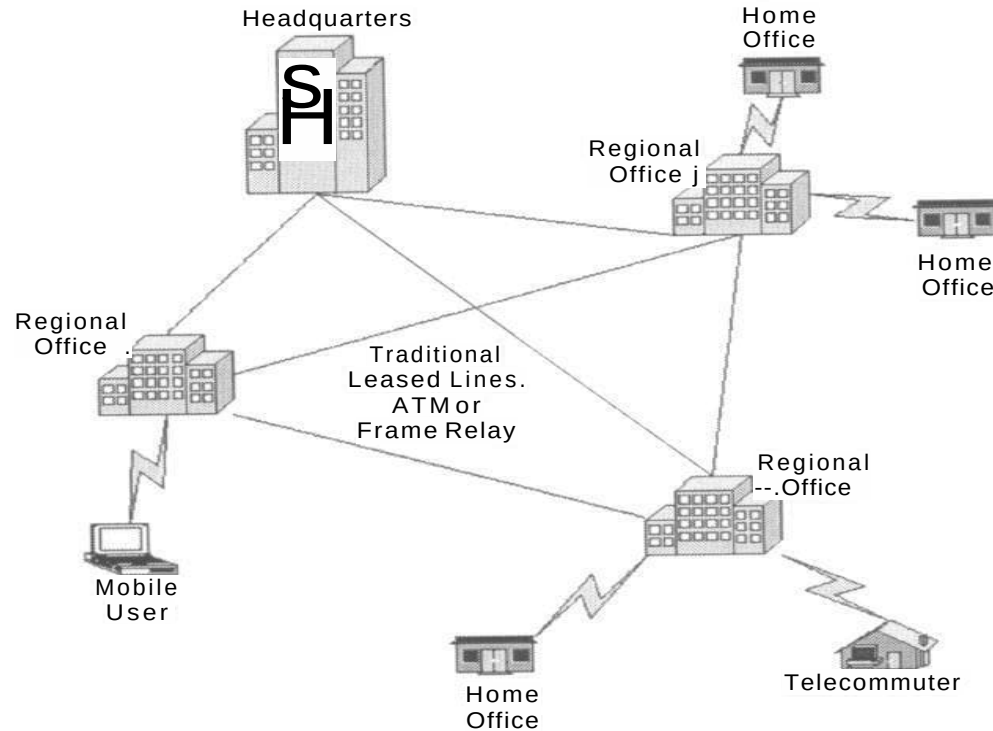
What is VPN?

- ✓ Virtual Private Network is a type of **private network that uses public telecommunication, such as the Internet**, instead of leased lines to communicate.
- ✓ Instead of using a dedicated connection, such as leased line, a **"virtual"** connection is made between various users and networks over a **shared or public network**, like the Internet.
- ✓ Data is transmitted as if it were passing through **private connections**.
- ✓ Became popular as more employees worked in **remote locations**.

What is VPN?

- In a virtual private network (VPN), "virtual" implies that there is no physical network infrastructure dedicated to the private network.
- Instead, a single physical network infrastructure is shared among various logical networks
- In VPNs, various networking technologies are applied toward the goal of providing private communications within the public Internet infrastructure.

Traditional Connectivity



Private Networks vs. Virtual Private Networks

- ✦ Employees can access the network (Intranet) from remote locations.
- ✦ Secured networks.
- ✦ The Internet is used as the backbone for VPNs
- ✦ Saves cost tremendously from reduction of equipment and maintenance costs.
- ✦ Scalability

Virtual Private Network (VPN) is basically of 2 types:

Remote Access VPN:

Remote Access VPN permits a user to connect to a private network and access all its services and resources remotely. The connection between the user and the private network occurs through the Internet and the connection is secure and private.

Remote Access VPN is useful for home users and business users both.

Site to Site VPN:

A Site-to-Site VPN is also called as Router-to-Router VPN and is commonly used in the large companies. Companies or organizations, with branch offices in different locations, use Site-to-site VPN to connect the network of one office location to the network at another office location.

Site to Site VPN:

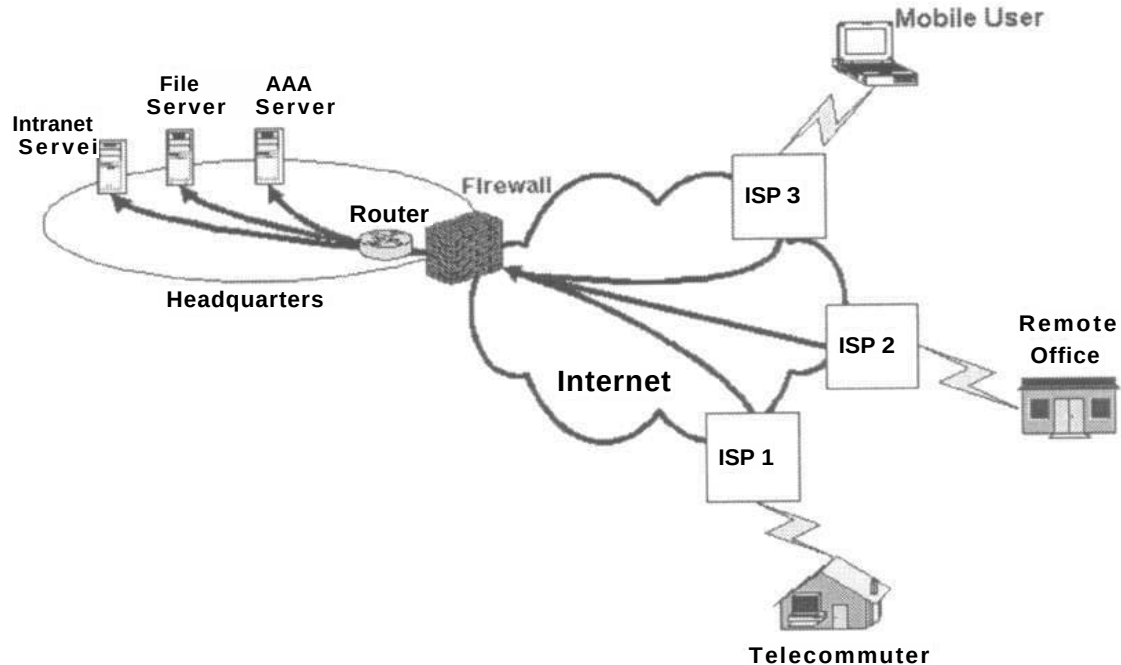
Intranet based VPN: When several offices of the same company are connected using Site-to-Site VPN type, it is called as Intranet based VPN.

Extranet based VPN: When companies use Site-to-site VPN type to connect to the office of another company, it is called as Extranet based VPN.

VPN Types

Use	Application	Alternative To	Benefits
Remote Access VPN	Remote Connectivity	Dedicated Dial ISDN	Ubiquitous Access LowerCost
IntranetVPN	Site-to-Site Internal Connectivity	Leased Line	Extend Connectivity LowerCost
ExtranetVPN	Business-to-Business External Connectivity	Fax, Mail, EDI	Facilitates E-Commerce

Remote Access Virtual Private Network



Brief Overview of How it Works

-
- ✓ Two connections - one is made to the Internet and the second is made to the VPN.
 - ✓ Firewalls - VPNs allow authorized users to pass through the firewalls.
 - ✓ Protocols - protocols create the VPN tunnels.

Types of Virtual Private Network (VPN) Protocols:

1. Internet Protocol Security (IPSec):
2. Layer 2 Tunneling Protocol (L2TP):
3. Point-to-Point Tunneling Protocol (PPTP):
4. SSL and TLS:
5. OpenVPN:
6. Secure Shell (SSH):

VPN Protocols

1. Internet Protocol Security (IPSec):

Internet Protocol Security, known as IPSec, is used to secure Internet communication across an IP network. IPSec secures Internet Protocol communication by verifying the session and encrypts each data packet during the connection.

IPSec runs in 2 modes:

- (i) Transport mode
- (ii) Tunneling mode

The work of transport mode is to encrypt the message in the data packet and the tunneling mode encrypts the whole data packet.

VPN Protocols

2. Layer 2 Tunneling Protocol (L2TP): L2TP or Layer 2 Tunneling Protocol is a tunneling protocol that is often combined with another VPN security protocol like IPsec to establish a highly secure VPN connection. L2TP generates a tunnel between two L2TP connection points and IPsec protocol encrypts the data and maintains secure communication between the tunnel.

3. Point-to-Point Tunneling Protocol (PPTP): PPTP or Point-to-Point Tunneling Protocol generates a tunnel and confines the data packet. Point-to-Point Protocol (PPP) is used to encrypt the data between the connection. PPTP is one of the most widely used VPN protocol and has been in use since the early release of Windows. PPTP is also used on Mac and Linux apart from Windows.

VPN Protocols

4. SSL and TLS:
SSL (Secure Sockets Layer) and TLS (Transport Layer Security) generate a VPN connection where the web browser acts as the client and user access is prohibited to specific applications instead of entire network. Online shopping websites commonly uses SSL and TLS protocol.

It is easy to switch to SSL by web browsers and with almost no action required from the user as web browsers come integrated with SSL and TLS.

SSL connections have "https" in the initial of the URL instead of "http".

VPN Protocols

5. OpenVPN:

OpenVPN is an open source VPN that is commonly used for creating Point-to-Point and Site-to-Site connections. It uses a traditional security protocol based on SSL and TLS protocol.

6. Secure Shell (SSH): Secure Shell or SSH generates the VPN tunnel through which the data transfer occurs and also ensures that the tunnel is encrypted. SSH connections are generated by a SSH client and data is transferred from a local port on to the remote server through the encrypted tunnel.

Difference Between Virtual Private Network (VPN) and Proxy

VPN:

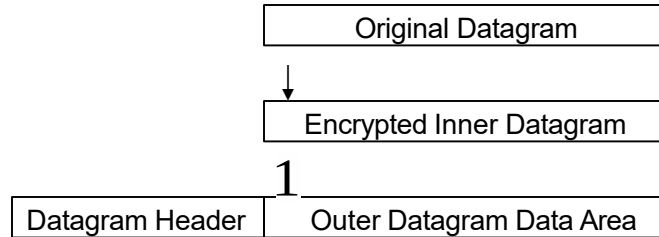
It stands for Virtual Private Network. It is a mechanism of employing **encryption, authentication and integrity protection** so that we can use public network as private network. It simulate a private network over public network. It allows users to remotely access a private network.

Proxy:

Actually Proxy server uses the anonymous network **id instead of actual IP address of client** (means it hides the IP address of client), so that the actual IP address of client couldn't be reveal.

Tunneling

A virtual point-to-point connection made through a public network.
It transports encapsulated datagrams.

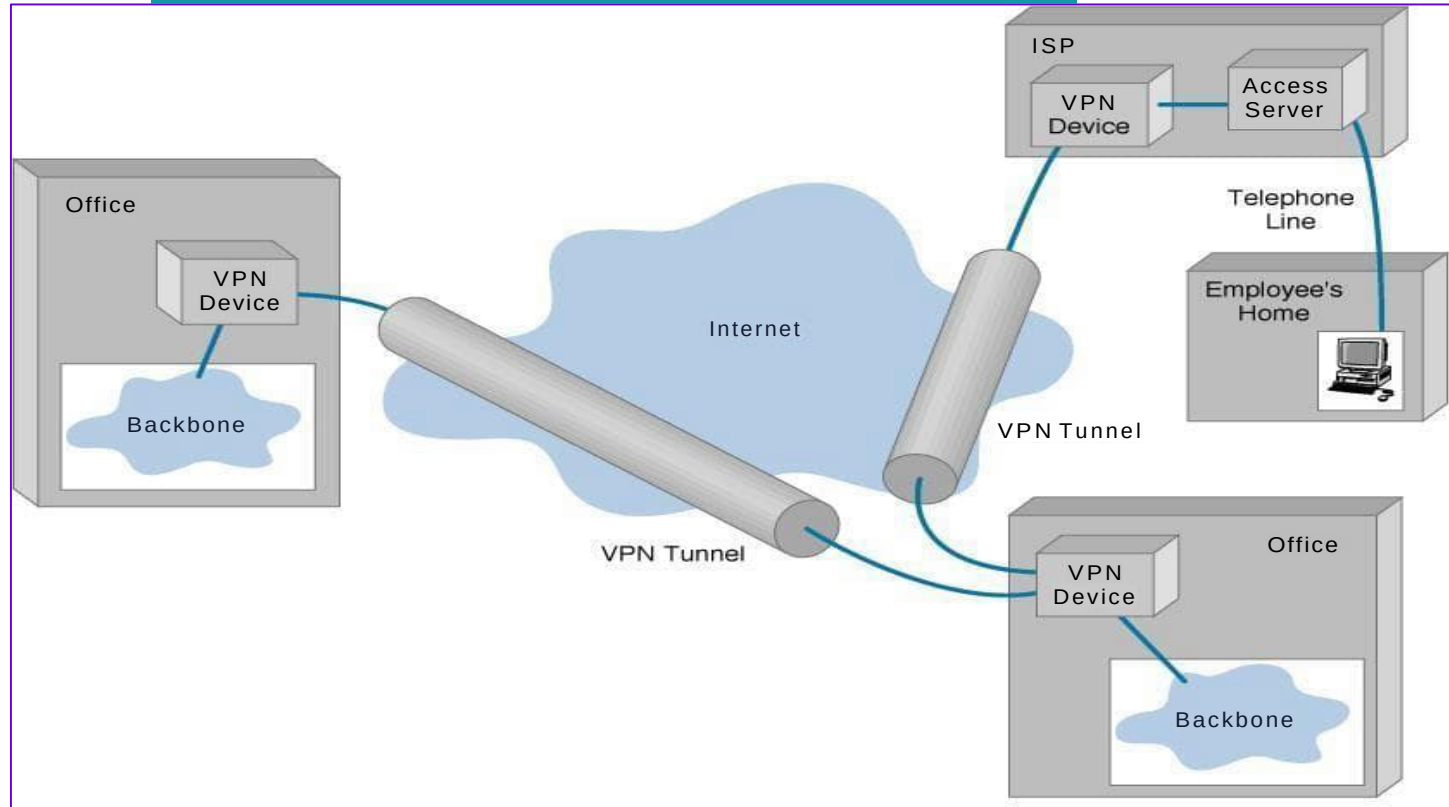


In computer networking, **Point-to-Point Protocol (PPP)** is a data link (layer 2) **protocol** used to establish a direct connection between two nodes.

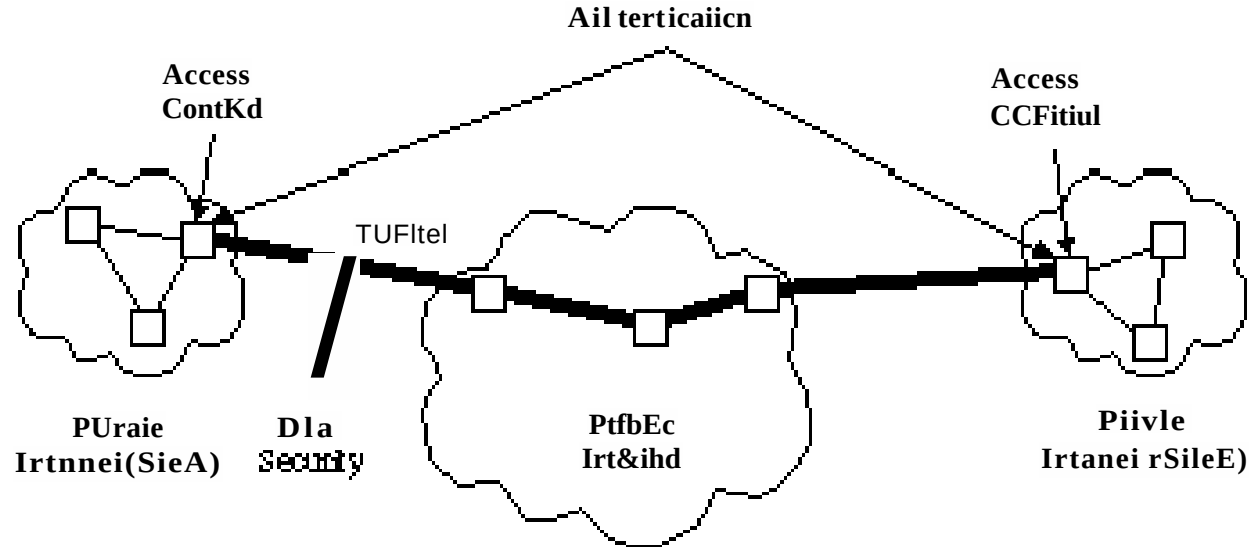
ATM layer regulates cells and cell transport and establishes and releases Virtual Circuits.

Virtual Private Networks (VPN)

Basic Architecture

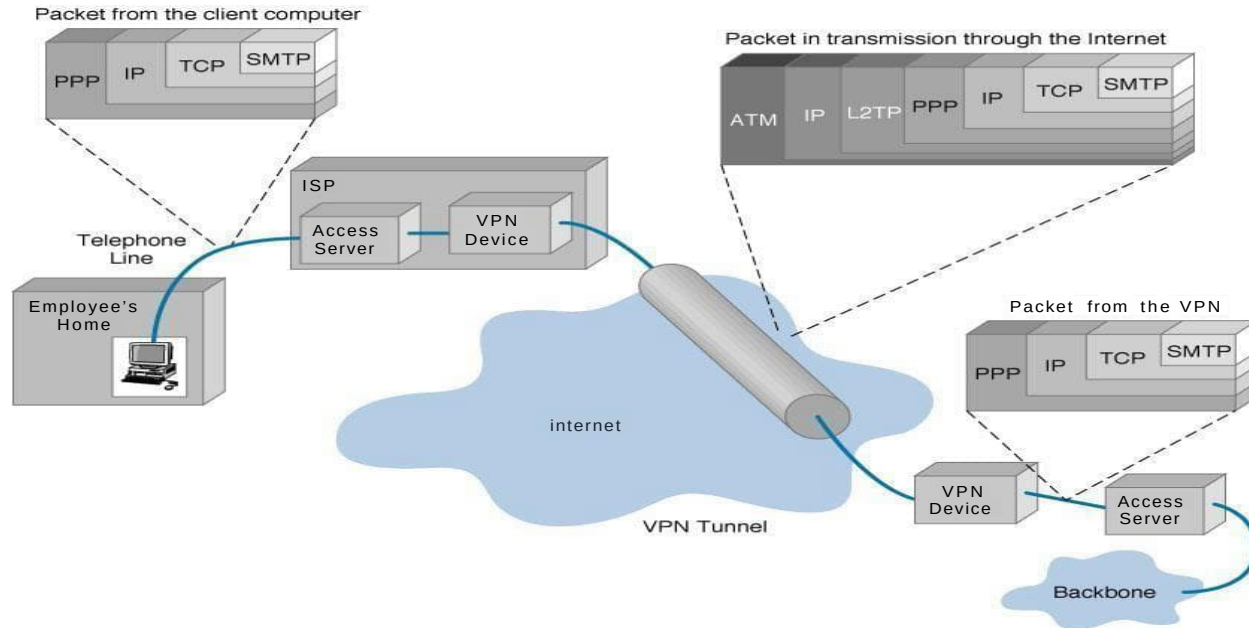


VPN Technologies



■ Tunneling – using encapsulation

VPN Encapsulation of Packets



VPN: Tunneling

Two Basic types of tunneling

- Site-to-Site
 - Typically uses GRE- Generic Routing Encapsulation.
- Remote-Access
 - Typically uses PPP
- **Note:**
 - **IPsec** stands for Internet Protocol Security while **GRE** stands for Generic Routing Encapsulation.
 - **IPsec** is the primary protocol of the Internet while **GRE** is not.
 - **GRE** can carry other routed protocols as well as IP packets in an IP network while **IPSec** cannot.

VPN Protocols (In depth)

Point-to-point tunneling protocol (PPTP)

- PPTP (Point-to-Point Tunnelling Protocol) is an OSI layer two protocols built on top of the PPP (Point-to-point protocol).
- PPP is a multi-protocol, dial-up protocol used to connect to the Internet.
- Remote users can access a private network via PPTP by first dialing into their local ISP.
- PPTP connects to the target network by creating a virtual network for each remote client.
- PPTP allows a PPP session, with non-TCP/IP protocols (e.g. IP, IPX or NetBEUI), to be tunneled through an IP network. PPTP is documented in RFC 2637 as an informational draft.

Layer Two tunneling protocol (L2TP) (In depth)

- L2TP (Layer 2 Tunnelling Protocol) is a combination of Microsoft PPTP (Point-to-Point Tunnelling Protocol) and Cisco L2F (Layer 2 Forwarding).
- L2TP can be used as a tunnelling protocol to encapsulate PPP (Point-to-Point Protocol) frames to be sent over IP, X.25, Frame Relay or ATM networks.
- Multiple connections are allowed through one tunnel. Like PPTP and L2F, L2TP operates on OSI layer two.
- Layer two VPN protocols encapsulate data in PPP frames and are capable of transmitting non-IP protocols over an IP network.

VPN Protocols (continued)

Internet Protocol Security Protocol (IPSec) provides enhanced security features such as ~~better encryption~~ algorithms and more comprehensive authentication.

IPSec has two encryption modes:

tunnel and **transport**. Tunnel encrypts the header and the payload of each packet while transport only encrypts the payload.

IPSec can encrypt data between various devices, such as:

- Router to router
- Firewall to router
- PC to router
- PC to server

What is WAP?

- A set of protocols which allow data exchange for mobile cellular systems
- The current world standard for the presentation and delivery of wireless information
- It is device independent
- It is network independent
- Wireless Application Protocol (**WAP**) is a technical standard for accessing information over a mobile wireless network.
- A **WAP** browser is a web browser for mobile devices such as mobile phones that uses the protocol.

Wireless Application Protocol (WAP)

- a universal, open standard developed to provide mobile wireless users access to telephony and information services
- have significant limitations of devices, networks, displays with wide variations
- WAP specification includes:
 - programming model, markup language, small browser, lightweight communications protocol stack, applications framework

Current Constraints of

Wireless Interfaces:

- Less Bandwidth
- High Latency
- Less Stable Connections
- Less Predictable Availability
- Diverse range of network standards

Mobile Devices:

Less CPU Power
Less Memory and Storage
Restricted Power Consumption
Small / Variable Sized Displays
Variable Input Types
(Keypad, Pen, etc,)

More Acronyms to learn

WAE - Wireless Application Environment

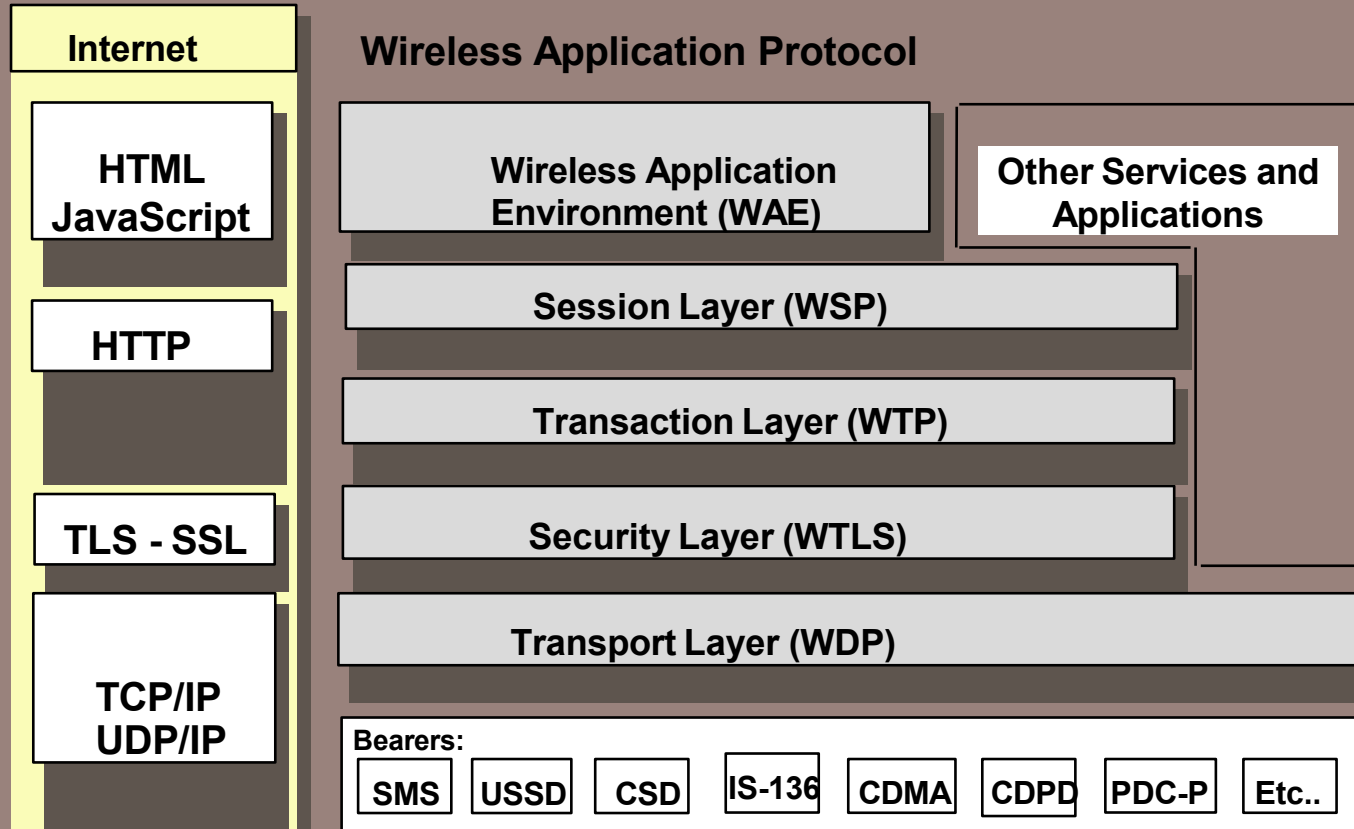
WSP - Wireless Session Protocol

WTP - Wireless Transaction Protocol

WTLS - Wireless Transport Layer Security

WDP - Wireless Datagram Protocol

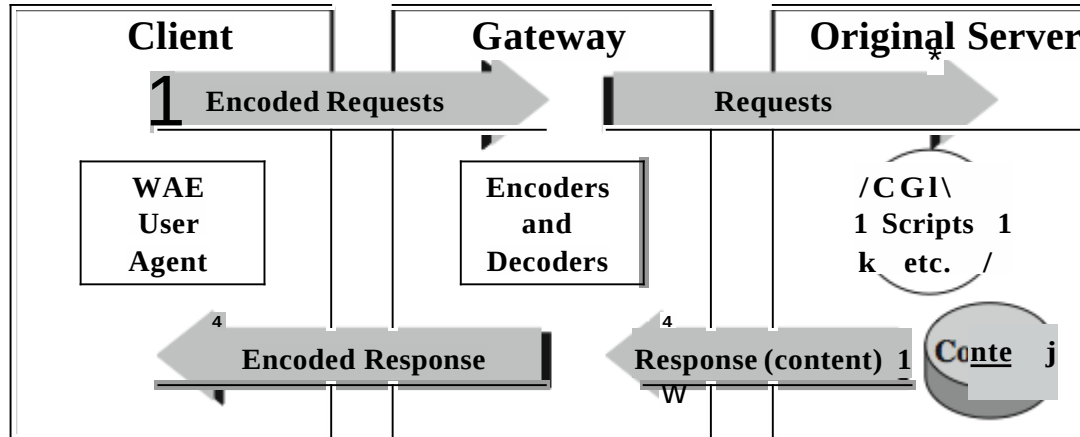
Comparison between Internet and WAP Models



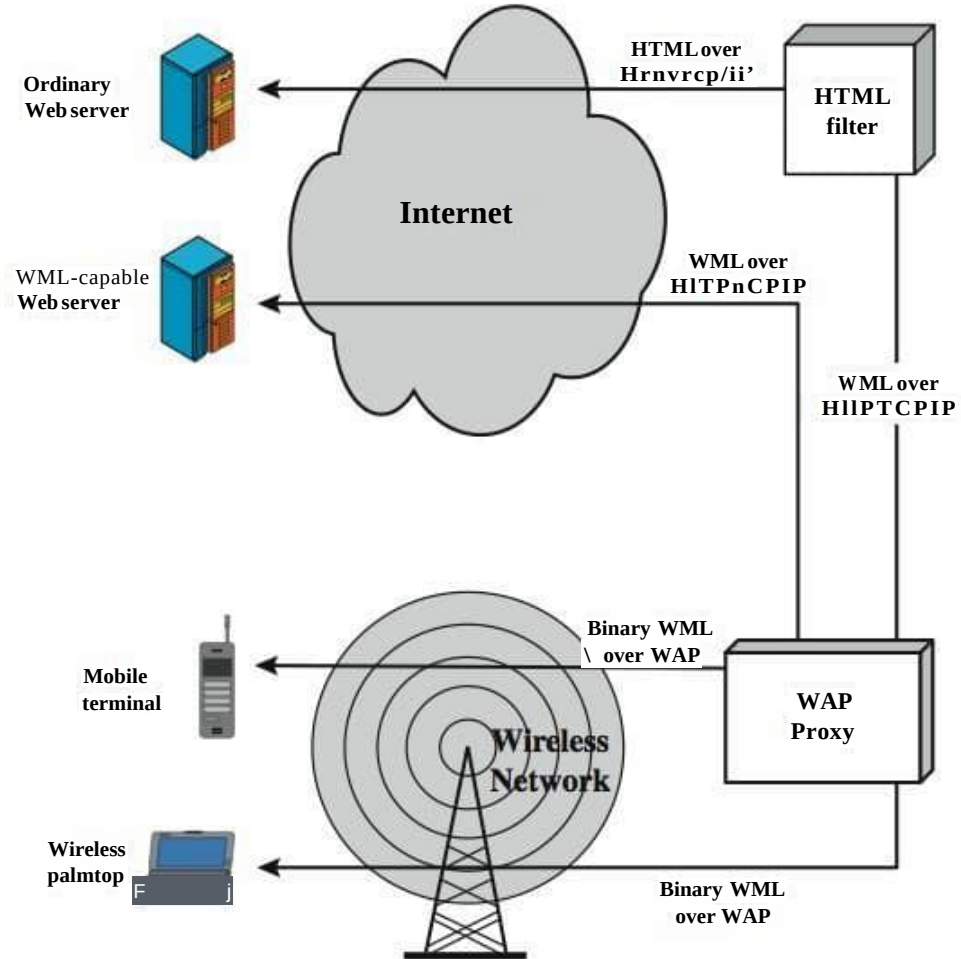
WAP Protocol Suite Contains

Wireless Application Environment (WAE)	WAP protocol suite
Wireless Session Protocol (WSP)	
Wireless Transaction Protocol (WTP)	
Wireless Transport Layer Security (WTLS)	
Wireless Datagram Protocol (WDP)	

WAP Programming Model



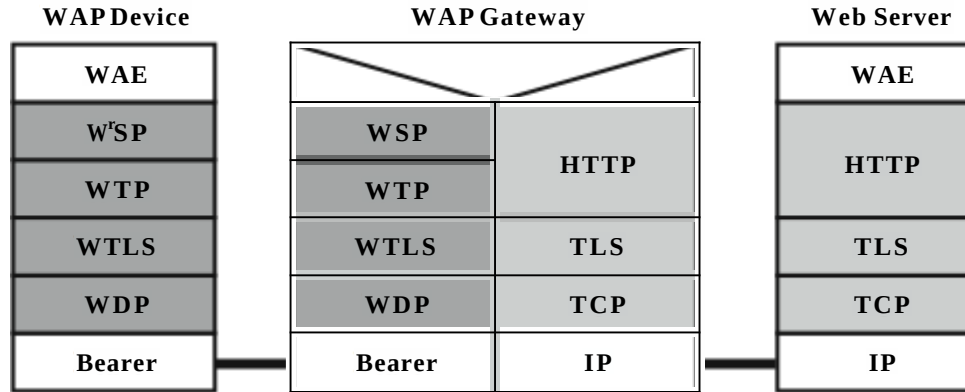
WAP Infra-structure



Wireless Markup Language

- describes content and format for data display on devices with limited bandwidth, screen size, and user input capability
- features include:
 - text / image formatting and layout commands
 - deck/card organizational metaphor
 - support for navigation among cards and decks
- a card is one or more units of interaction
- a deck is similar to an HTML page

WTP Gateway



WAP Protocols

- Wireless Session Protocol (WSP)
 - provides applications two session services
 - connection-oriented and connectionless
 - based on HTTP with optimizations
- Wireless Transaction Protocol (WTP)
 - manages transactions of requests / responses between a user agent & an application server
 - provides an efficient reliable transport service
- Wireless Datagram Protocol (WDP)
 - adapts higher-layer WAP protocol to comms

Wireless Transport Layer Security (WTLS)

- provides security services between mobile device (client) and WAP gateway
 - provides data integrity, privacy, authentication, denial-of-service protection
- based on TLS
 - more efficient with fewer message exchanges
 - use WTLS between the client and gateway
 - use TLS between gateway and target server
- WAP gateway translates WTLS / TLS

WTLS Sessions and Connections

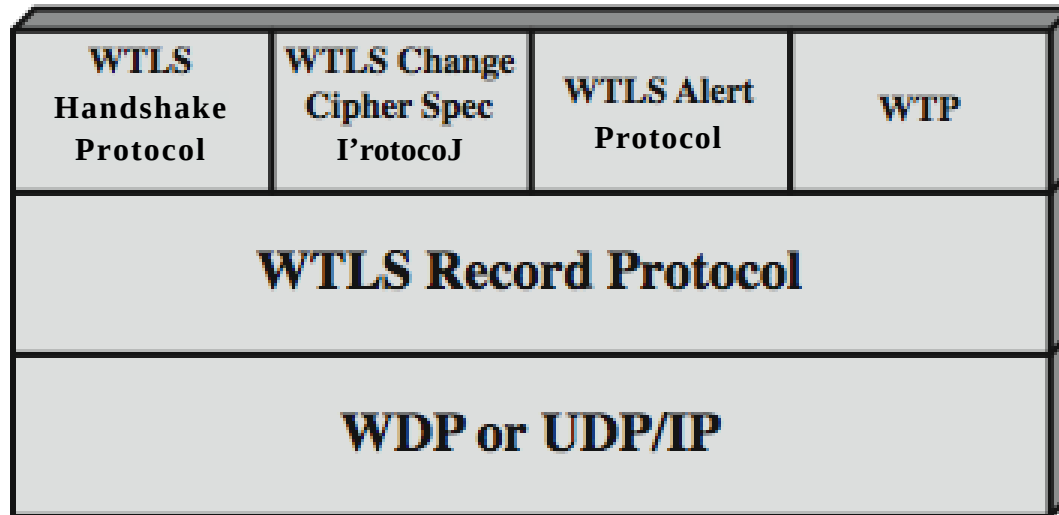
➤ **secure connection**

- a transport providing a suitable type of service
- connections are transient
- every connection is associated with 1 session

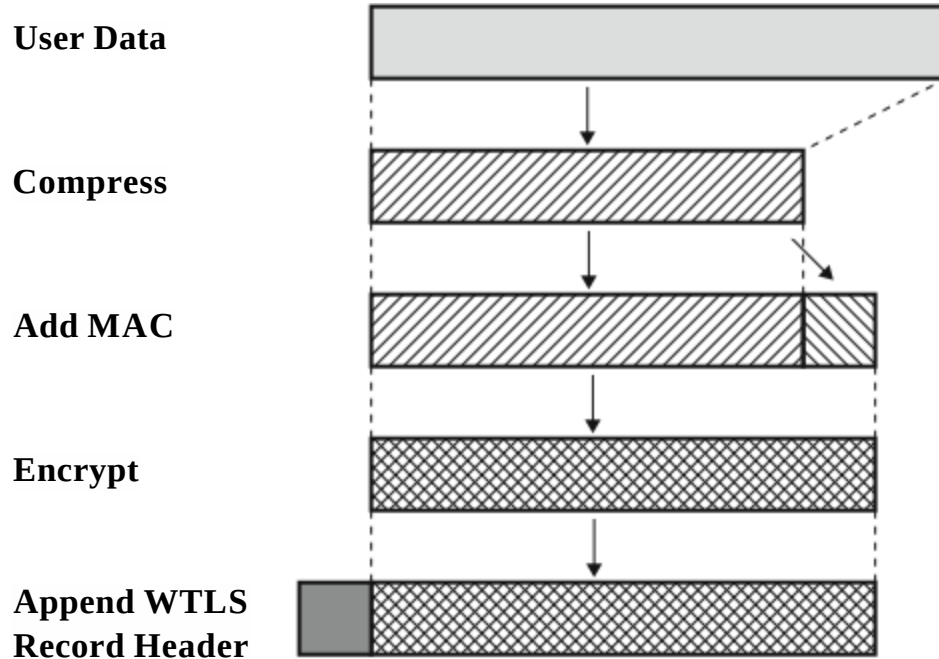
➤ **secure session**

- an association between a client and a server
- created by Handshake Protocol
- define set of cryptographic security parameters
- shared among multiple connections

WTLS Protocol Architecture



WTLS Record Protocol

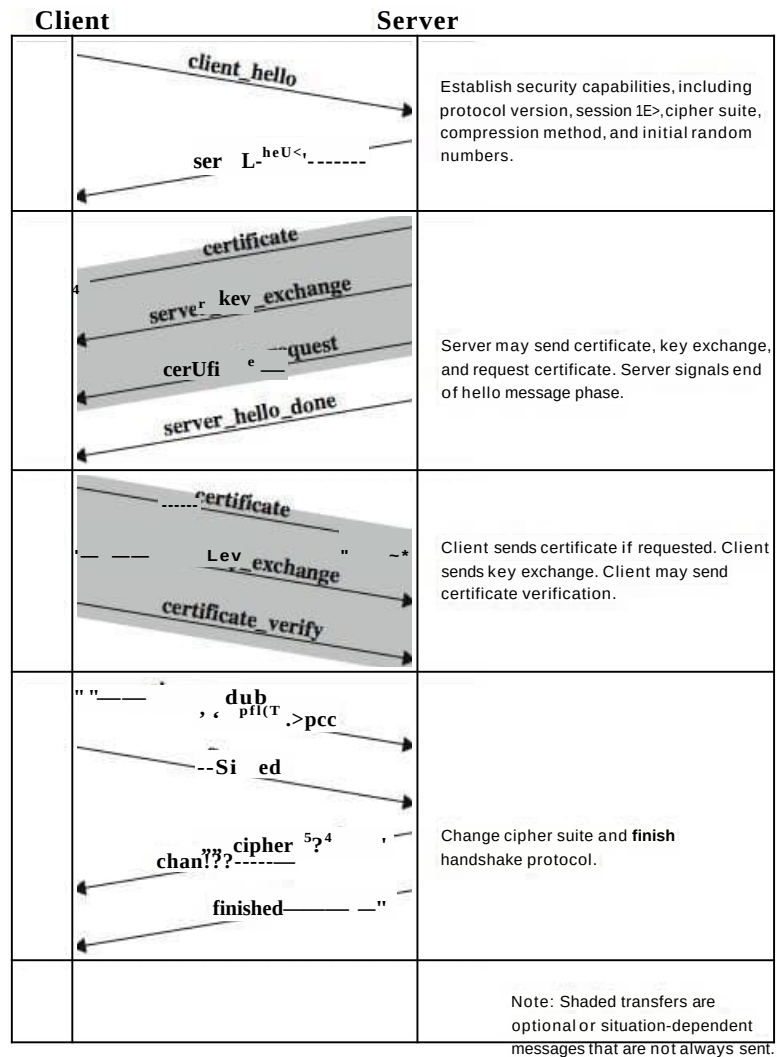


WTLS Higher-Layer Protocols

- Change Cipher Spec Protocol
 - simplest, to make pending state current
- Alert Protocol
 - used to convey WTLS-related alerts to peer
 - has severity: warning, critical, or fatal
 - and specific alert type
- Handshake Protocol
 - allow server & client to mutually authenticate
 - negotiate encryption & MAC algs & keys

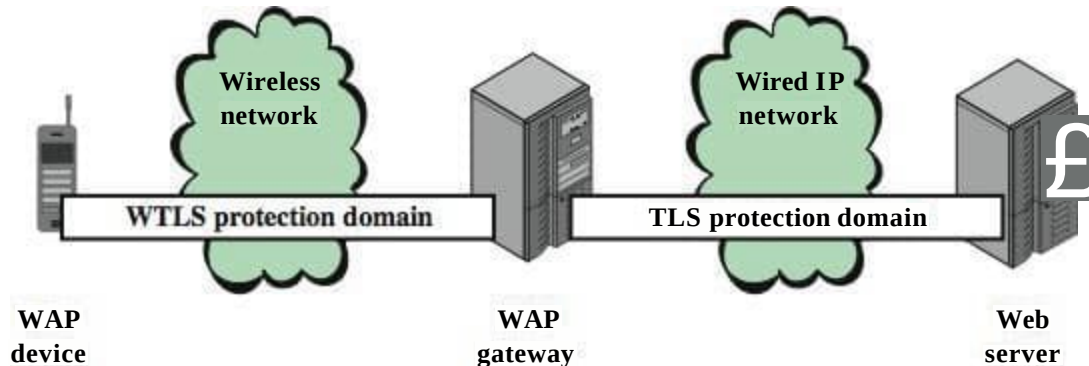
Handshake Protocol

Time ↓

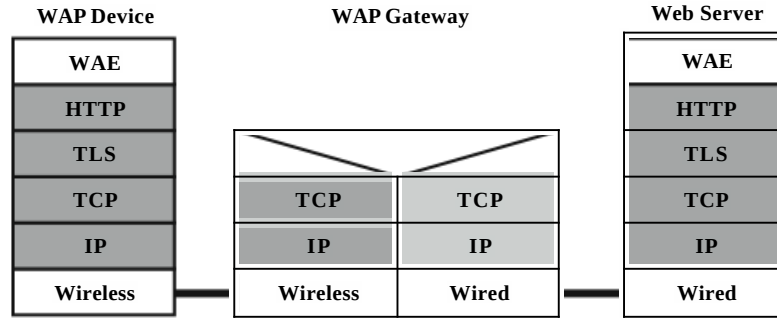


WAP End-to-End Security

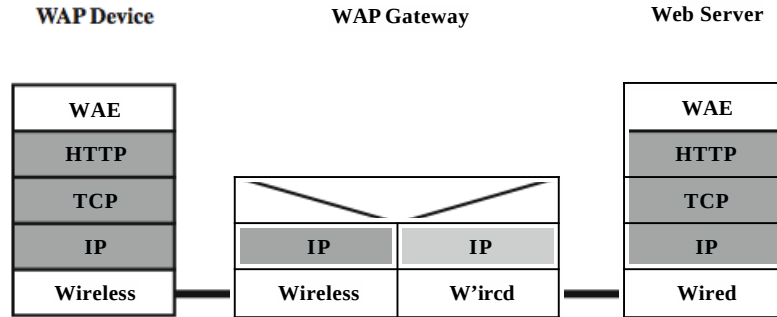
- have security gap end-to-end
 - at gateway between WTLS & TLS domains



WAP2 End-to-End Security

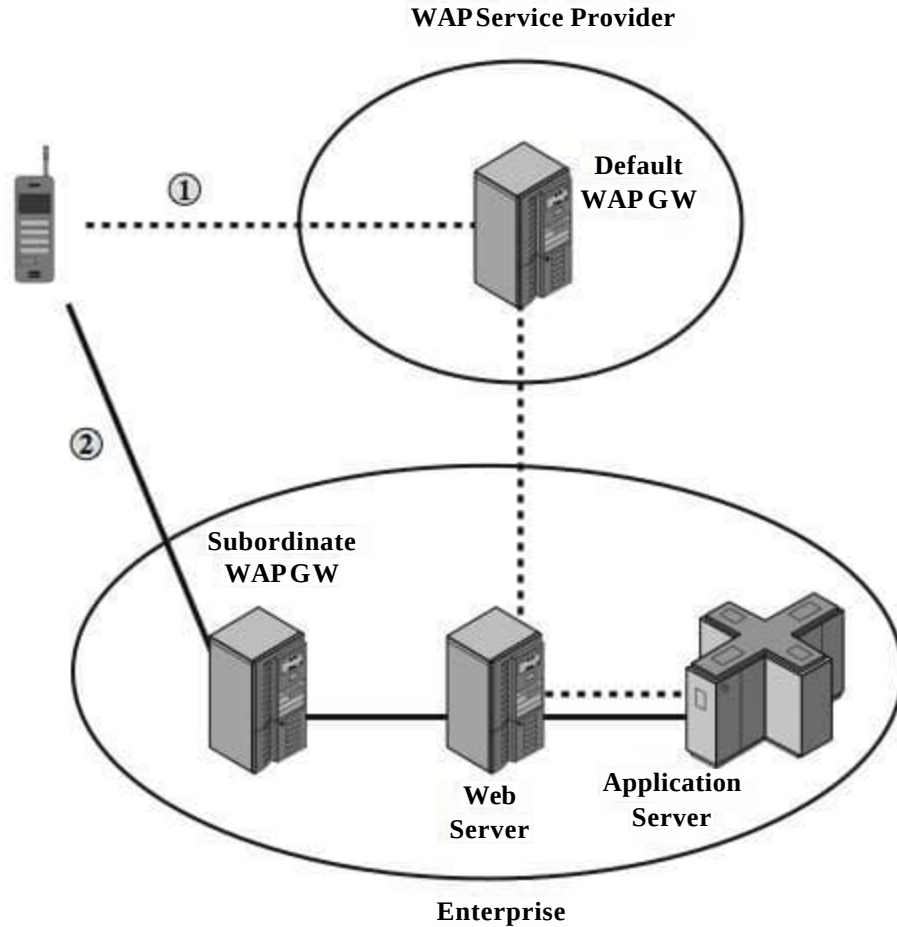


(a) TLS-based security



(b) IPSec-based security

WAP2 End-to-End Security



Electronic mail security

Pretty Good Privacy

- Philip R. Zimmerman is the creator of PGP.
- PGP provides a confidentiality and authentication service that can be used for electronic mail and file storage applications.

Operational Description

➤ Consist of five services:

- Authentication
- Confidentiality
- Compression
- E-mail compatibility
- Segmentation

Summary of PGP Services

Function	Algorithms Used	Description
Digital signature	DSS/SHA or RSA/SHA	A hash code of a message is created using SHA-1. This message digest is encrypted using DSS or RSA with the sender's private key and included with the message.
Message encryption	CAST or IDEA or Three-key Triple DES with Diffie-Hellman or RSA	A message is encrypted using CAST-128 or IDEA or 3DES with a one-time session key generated by the sender. The session key is encrypted using Diffie-Hellman or RSA with the recipient's public key and included with the message.
Compression	ZIP	A message may be compressed for storage or transmission using ZIP.
E-mail compatibility	Radix-64 conversion	To provide transparency for e-mail applications, an encrypted message may be converted to an ASCII string using radix-64 conversion.

Compression

- PGP compresses the message after applying the signature but before encryption.
- The placement of the compression algorithm is critical.
- The compression algorithm used is ZIP.

E-mail Compatibility

- The scheme used is radix-64 conversion.
- The use of radix-64 expands the message by 33%.

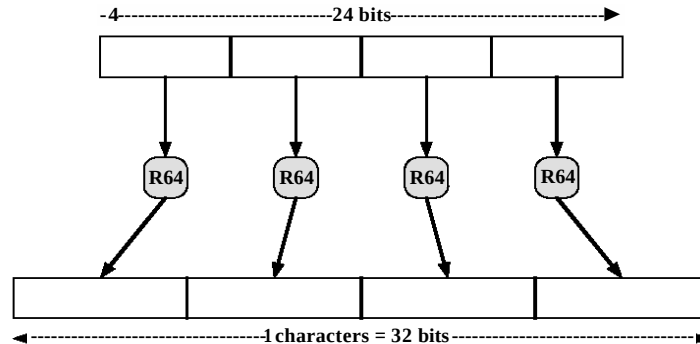


Figure 5.11 Printable Encoding of Binary Data into Radix-64 Format
Henric Johnson

Summary of PGP Services

Function	Algorithm Used
Digital Signature	DSS/SHA or RSA/SHA
Message	CAST or IDEA or three-key
Encryption	triple DES with Diffie-Hellman or RSA
Compression	ZIP
E-mail	Radix-64 conversion
Compatibility	

S/MIME

- Secure/Multipurpose Internet Mail Extension
- S/MIME will probably emerge as the industry standard.
- PGP for personal e-mail security

Algorithms Used

- **Message Digesting:** SHA-1 and MDS
- **Digital Signatures:** DSS
- **Secret-Key Encryption:** Triple-DES, RC2/40 (exportable)
- **Public-Private Key Encryption:** RSA with key sizes of 512 and 1024 bits, and Diffie-Hellman (for session keys).

What is SMTP and which are the default SMTP ports

- Simple Mail Transfer Protocol (SMTP) is the standard protocol for **sending emails** across the Internet.
- By default, the SMTP protocol works on three ports:
- **Port 25** - this is the default **SMTP non-encrypted port**
- **Port 2525** - this port is opened on all SiteGround servers in case port 25 is filtered (by your ISP for example) and **you want to send non-encrypted emails with SMTP.**
- **Port 465** - this is the port used if you want to send messages using **SMTP securely.**

What is POP3 and which are the default POP3 ports

- Post Office Protocol version 3 (POP3) is a standard mail protocol used to **receive emails from a remote server to a local email client**.
- POP3 allows you to download email messages on your local computer and read them even when you are offline.
- **Note, that when you use POP3 to connect to your email account, messages are downloaded locally and removed from the email server.**
- By default, the POP3 protocol works on two ports:
- **Port 110** - this is the default POP3 non-encrypted port
- **Port 995** - this is the port you need to use if you want to connect using POP3 securely

The Internet Message Access Protocol (IMAP)

- IMAP is a mail protocol used for accessing email on a **remote web server from a local client**.
- **IMAP and POP3** are the two most commonly used Internet mail protocols for **retrieving emails**.
- Both protocols are supported by all modern email **clients and web servers**.
- While the **POP3 protocol assumes that your email is** being accessed only from **one application**,
- IMAP allows simultaneous **access by multiple clients**.
- This is why IMAP is more suitable for you if you're going to access your email from **different locations or if your messages are managed by multiple users**.
- By default, the IMAP protocol works on two ports:
- **Port 143** - this is the default IMAP **non-encrypted port**
- **Port 993** - this is the port you need to use if you want to connect using **IMAP securely**

Thank you

